OFFICIAL STANDARD OPERATING MANUAL

WarehousePOS System Operations Guide

Complete User Guide Featuring Actual System Screenshots and Screen-by-Screen Navigation Instructions for Point of Sale, Warehouse Inventory, Invoicing, Procurement, and Financial Analytics

- ✓ Actual WPF Live System Interface
- ✓ 100% Offline Single-PC Architecture
- ✓ Epson LQ-310 Dot-Matrix Printing
- ✓ Multi-Role Security & Permissions
- ✓ Step-by-Step Screen Navigation

SYSTEM EDITION Desktop v1.0 (Windows x64)	HARDWARE PROFILE Epson LQ-310 Continuous 9.5"×5.5"	BUSINESS NAME Happy Products
DATABASE ENGINE SQLite (Embedded EF Core 10)	DEPLOYMENT TYPE 100% Offline Standalone Desktop	DOCUMENTATION DATE October 2026

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01 System Overview & Local Directories

WarehousePOS is designed as a high-performance, single-PC offline Windows application built on **C# (.NET 10)**, **WPF**, and **Entity Framework Core with SQLite**. It powers all retail transactions, warehouse inventory operations, and dot-matrix continuous invoice printing for **Happy Products** without requiring cloud servers or active internet connections.

1.1 Operating Environment & Storage Paths

Resource	Local Windows Path	Function & Security
Live Database	`C:\ProgramData\WarehousePOS\Data\WarehousePOS.db`	Permanent operational store (WAL mode enabled).
Automatic Backups	`C:\ProgramData\WarehousePOS\Backups\`	Daily snapshots taken automatically on launch.
System Logs	`C:\ProgramData\WarehousePOS\Logs\application.log`	Serilog structured logs (30-day retention).
Application Binaries	`C:\Program Files\WarehousePOS\WarehousePOS.Desktop.exe`	64-bit self-contained desktop binary.

02 User Authentication (Actual System Screen)

System access is secured using individual credentials. Passwords are never stored in plain text; all authentication utilizes salted **BCrypt** hashing.

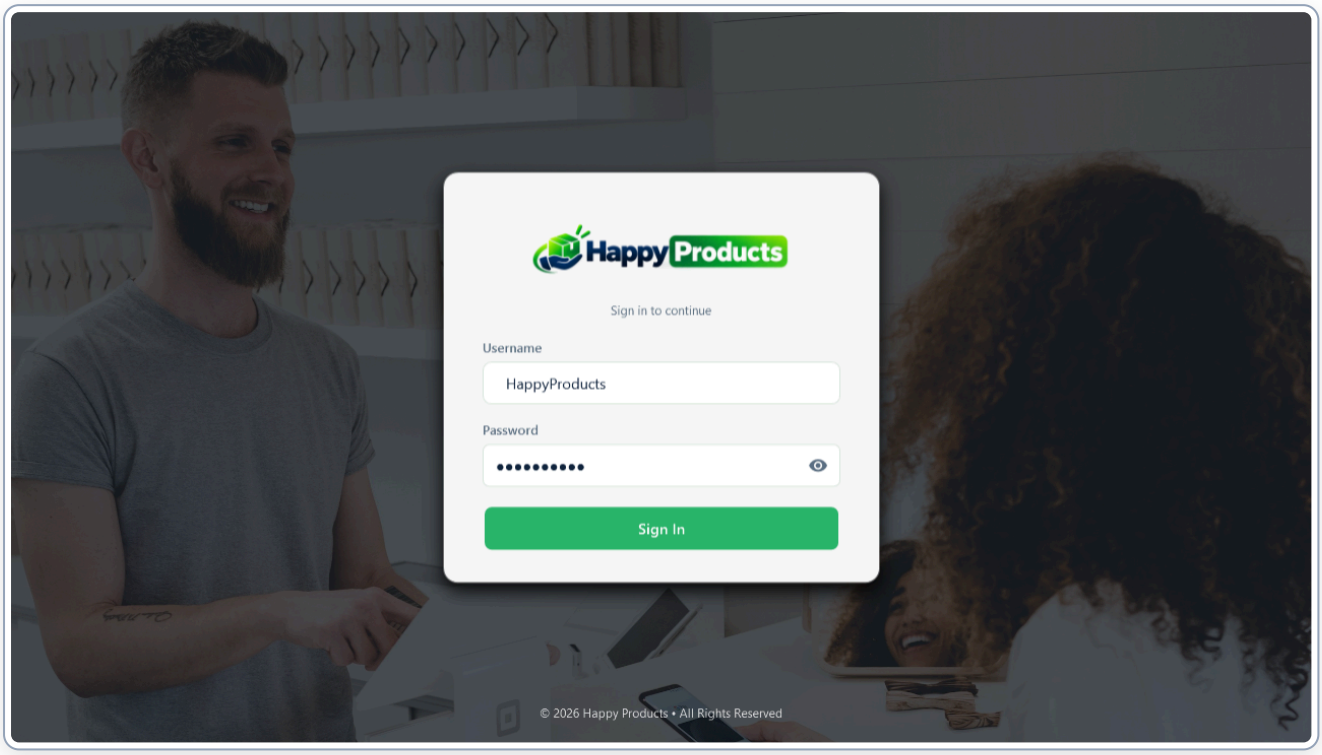


Figure 2.1: Actual Happy Products Sign-In Window displaying entry of sample demonstration credentials.

2.1 How to Sign In with Sample Data

1. **Enter Username:** Click into the *Username* text box. Type the assigned handle. For demonstration and training, use sample account: `HappyProducts` (or cashier accounts `cashier1`, `cashier2`).
2. **Enter Password:** Click into the *Password* box. The input displays security bullets (e.g., `••••••••••`). Use the eye icon toggle button on the right edge to briefly verify input if needed.
3. **Sign In:** Click the green **Sign In** button or press `Enter`. The authentication event is recorded in the `AuditLog` and the system transitions directly into the main shell.

03 Application Navigation Architecture

WarehousePOS features a persistent left navigation sidebar. It provides clear visual cues indicating the active view with a green highlight and left accent bar.

3.1 Sidebar Navigation Buttons & Permissions

Sidebar Button	Target View	Required Role	Keyboard Shortcut
🖥️ **Point of Sale**	Point of Sale (Cash) Terminal	WORKER / ADMIN	Ctrl + 1 or Default Landing
🧾 **Bill History**	Bill History & Invoices	WORKER / ADMIN	Ctrl + 2
📦 **Products**	Product Catalog & Stock Levels	WORKER / ADMIN	Ctrl + 3
🛒 **Purchasing**	Purchasing & Stock Inbound (GRN)	ADMIN / Manager	Ctrl + 4
🚚 **Suppliers**	Suppliers Directory & Balances	ADMIN / Manager	Ctrl + 5
👥 **Customers**	Customer Directory & Credit	WORKER / ADMIN	Ctrl + 6
📊 **Reports**	Executive Analytics & Reports	ADMIN ONLY	Ctrl + 7
💸 **Expenses**	Operational Expenses Tracking	ADMIN ONLY	Ctrl + 8
🔐 **User Management**	User Accounts & Payroll Setup	ADMIN ONLY	Ctrl + 9
⚙️ **Settings**	Store Information & Hardware	ADMIN ONLY	Ctrl + 0
⬅ **Logout**	Returns to Login Window	All Users	Alt + L

04 Point of Sale (Cash) Terminal

NAVIGATION Click 🖥️ **Point of Sale** on the sidebar, or press `F1` from any screen.

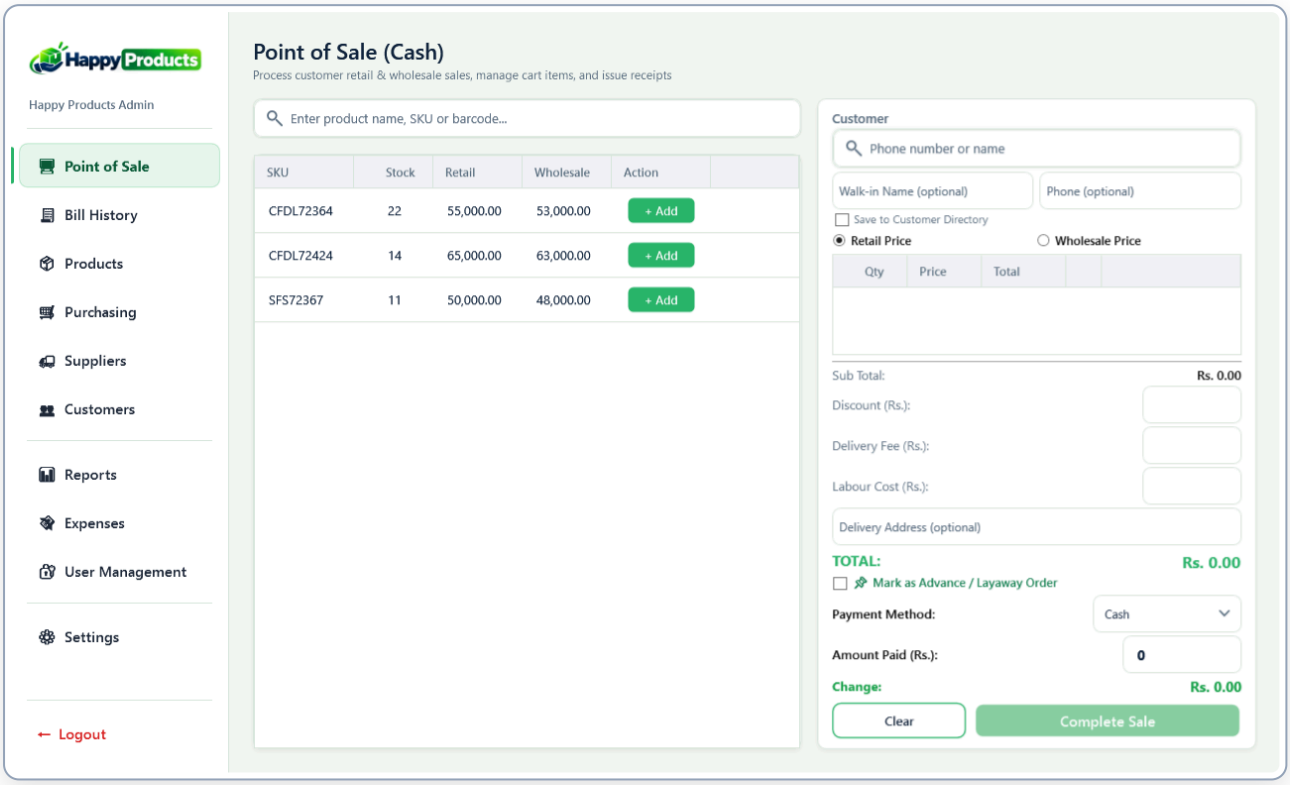


Figure 4.1: Actual Point of Sale (Cash) Terminal showing live catalog, active price tiers, customer input, and billing panel.

4.1 On-Screen Controls & Checkout Workflow

1. **Product Search Bar:** Type product name, SKU (e.g., `CFDL72364`), or scan the barcode. Items appear in the table with stock counts and prices.
2. **Catalog Quick Add:** Click the green **+ Add** button on any item row to add 1 unit to the cart.
3. **Customer Assignment:**
   - For counter sales, leave as Walk-in Customer.
   - To record a client name/phone, type in *Walk-in Name* and *Phone*.
   - Check **Save to Customer Directory** to automatically register them for credit tracking.
4. **Price Tier Selection:** Switch between **Retail Price** and **Wholesale Price** using the radio buttons.
5. **Concessions & Surcharges:** Enter optional *Discount (Rs.)*, *Delivery Fee (Rs.)*, or *Labour Cost (Rs.)*. The TOTAL updates dynamically.
6. **Payment Settlement:** Select *Cash*, enter *Amount Paid (Rs.)*. The exact change calculates instantly. Click **Complete Sale** (or press `F12`) to finalize the transaction, record inventory decrement, and trigger the Epson LQ-310 printer.

05 Bill History & Invoice Inquiries

NAVIGATION Click **Bill History** on the left navigation sidebar.

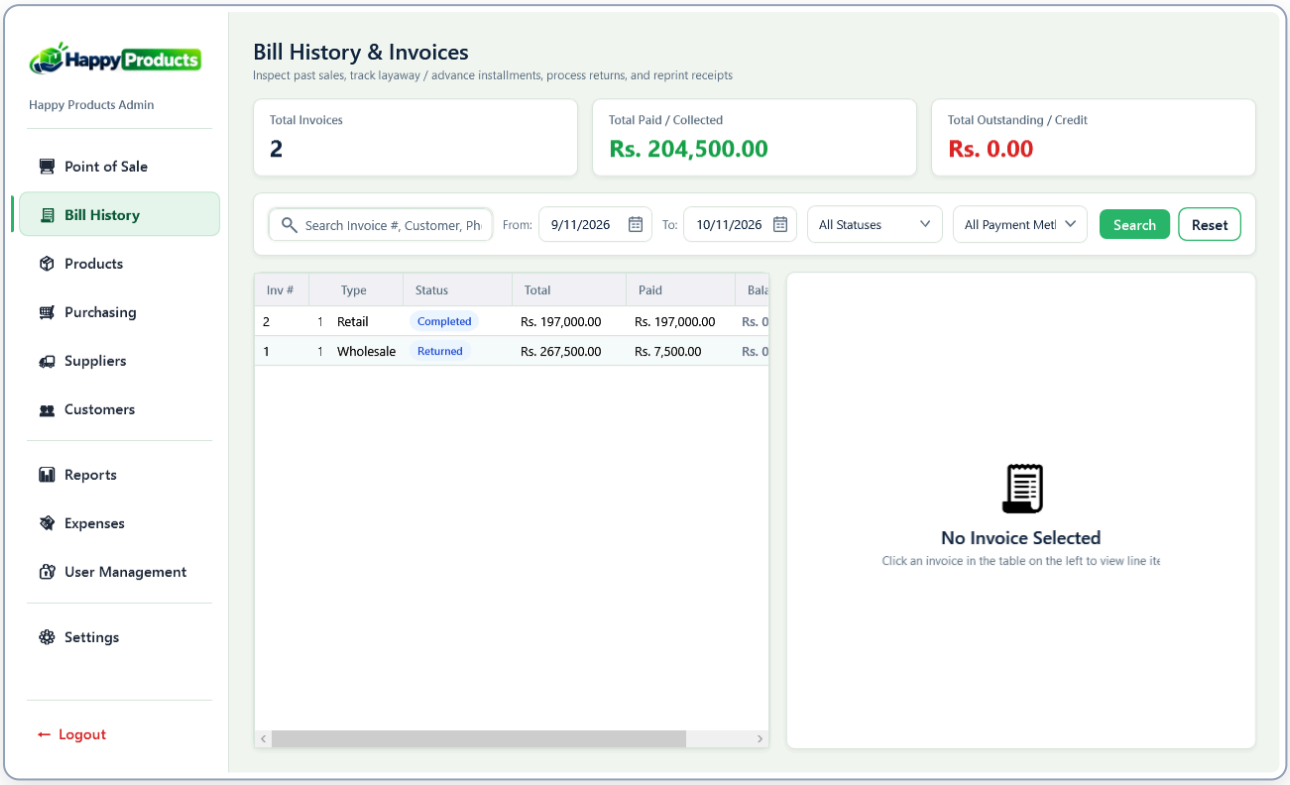


Figure 5.1: Actual Bill History & Invoices screen displaying historical invoices, totals, statuses, and search filters.

5.1 On-Screen Features & Actions

- **Summary Metrics:** Shows *Total Invoices*, *Total Paid / Collected (Rs.)*, and *Total Outstanding / Credit (Rs.)* for the selected date window.
- **Search & Filter Bar:** Filter by invoice number, customer phone number, date range (From / To), Status (*Completed*, *Returned*, *Layaway*), and Payment Method. Click **Search** to query.
- **Invoice Details Viewer:** Click any invoice row in the table to display the itemized breakdown on the right pane, inspect discounts, verify cashier name, and click **Reprint Receipt** to reissue physical bills.
- **Returns & Adjustments:** Select an eligible invoice to process customer product returns or adjust delivery expenses.

06 Receipt Generation & Epson LQ-310 Printing

Happy Products uses an **Epson LQ-310 24-pin Dot-Matrix Printer** running continuous tractor-feed 9.5" × 5.5" (half-page) 2-ply NCR paper.

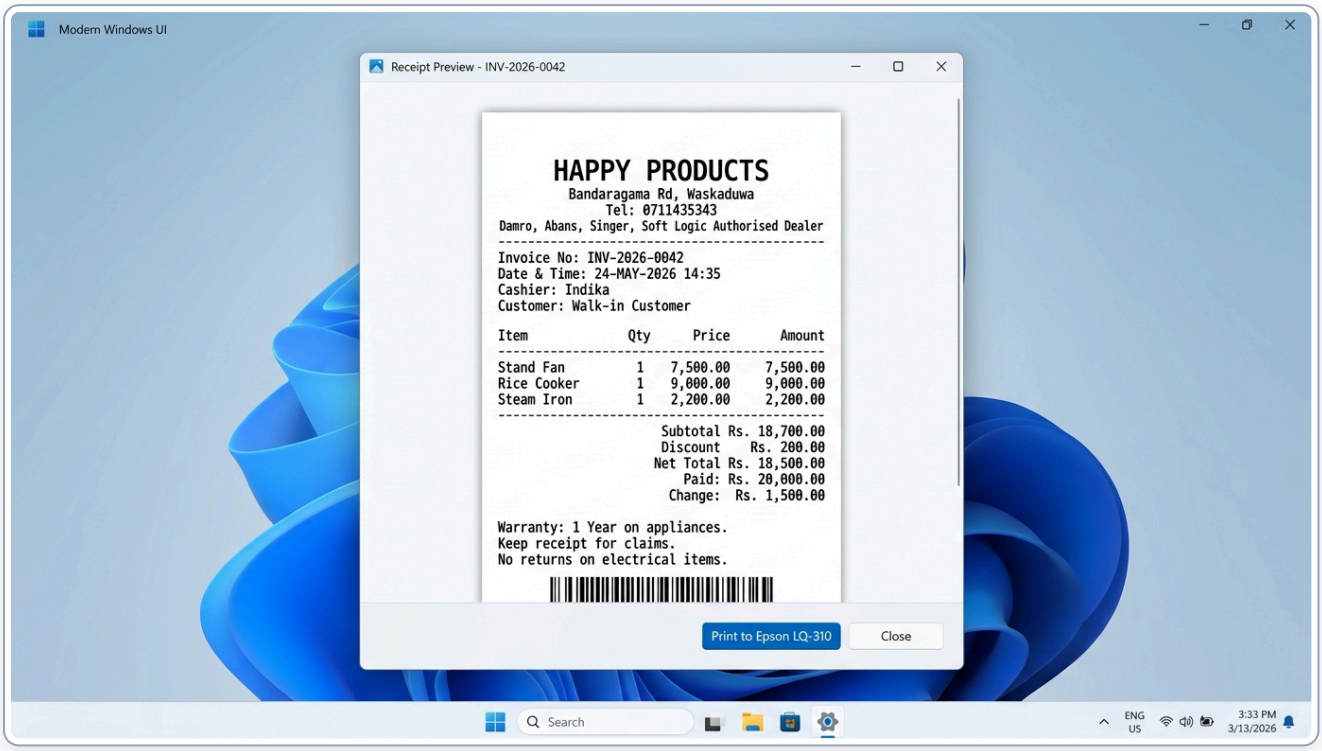


Figure 6.1: Dot-Matrix Receipt Preview showing store header, itemized lines, totals, warranty policy, and optical barcode.

6.1 Key Receipt Elements

- **Store Header:** HAPPY PRODUCTS • Bandaragama Rd, Waskaduwa • Tel: 0711435343 • Damro, Abans, Singer, Soft Logic Authorised Dealer.
- **Metadata:** Sequential invoice ID (e.g., `INV-2026-0042`), timestamp, cashier name, customer name.
- **Line Items:** Description, quantity, unit price (Rs.), and extended line total.
- **Financials:** Subtotal, discounts, net payable, cash tendered, and change returned.
- **Warranty Terms:** Clear warranty policy disclaimer stating manufacturing repair depot terms and inspection guidelines.
- **Optical Barcode:** Code128 barcode of invoice number allowing instant scanning for returns and warranty claims.

07 Product Catalog & Inventory Management

NAVIGATION Click 📦 **Products** on the left navigation sidebar.

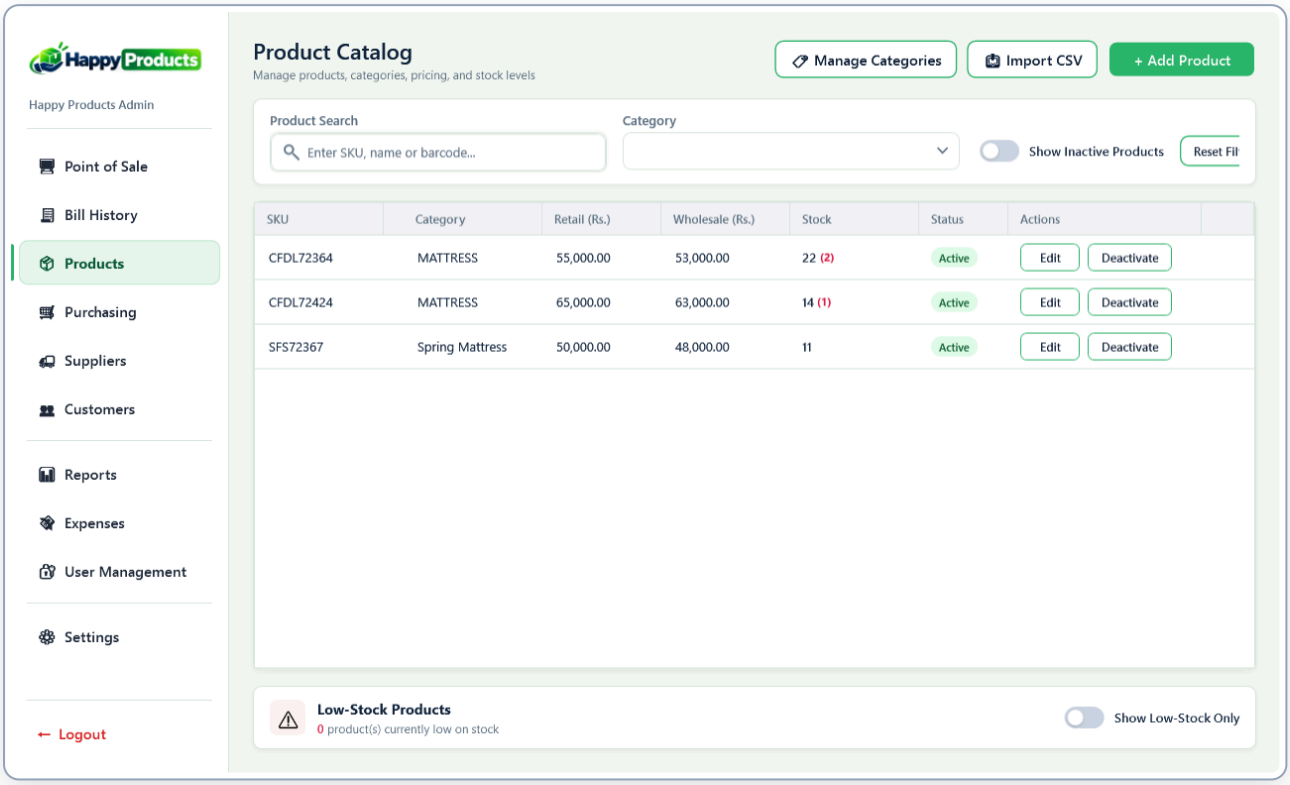


Figure 7.1: Actual Product Catalog screen displaying master inventory records, stock levels, categories, and actions.

7.1 On-Screen Actions & Workflows

- **Add New Product:** Click the green **+ Add Product** button in the top right to open the product registration form (SKU, barcode, category, cost price, retail price, wholesale price, and reorder levels).
- **Manage Categories:** Click **Manage Categories** to create and organize product categories (e.g., MATTRESS, Home Appliances, Electronics).
- **Bulk CSV Import:** Click **Import CSV** to batch-upload hundreds of product records from a spreadsheet.
- **Stock Level Tracking:** The *Stock* column highlights available physical units and displays low-stock indicators in red text.
- **Low-Stock Products Alert:** The bottom banner shows current low-stock count and includes a *Show Low-Stock Only* toggle switch for rapid reordering.
- **Editing & Deactivation:** Use **Edit** to modify prices or specifications; use **Deactivate** for safe soft-deletion without breaking historical invoice data.

08 Purchasing & Stock Inbound (GRN)

NAVIGATION Click 🛒 **Purchasing** on the left navigation sidebar (Admin / Manager role required).

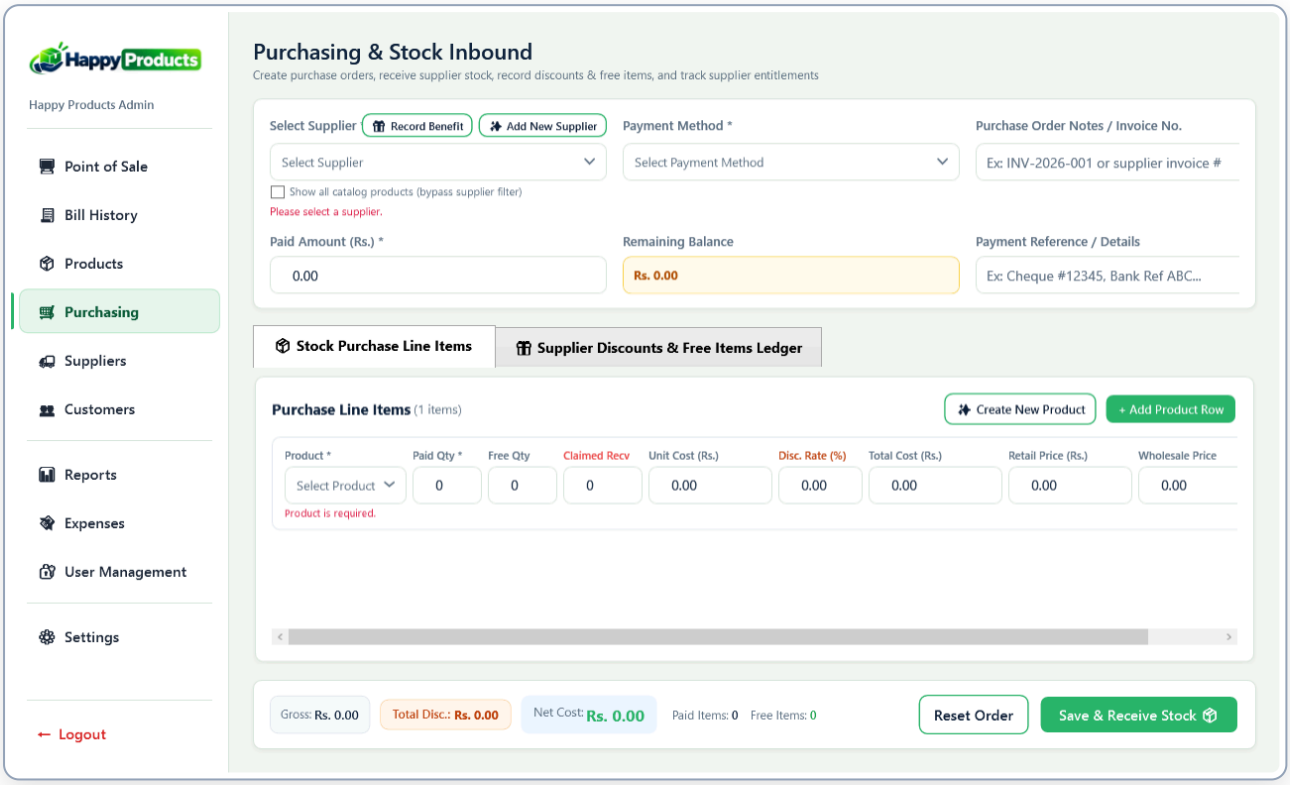


Figure 8.1: Actual Purchasing & Stock Inbound view for processing supplier purchase orders, discounts, and GRN receipts.

8.1 Procurement & Goods Receiving Workflow

1. **Supplier & Terms:** Select the vendor from the *Select Supplier* dropdown (e.g., Arpico, Piyestra Furniture). Choose payment method (Cash, Bank Transfer, Cheque, Credit).
2. **Invoice Reference:** Type supplier delivery note or invoice number in *Purchase Order Notes / Invoice No.*
3. **Add Purchase Items:** Select product, input *Paid Qty*, *Free Qty* (bonus units), *Unit Cost (Rs.)*, and *Discount Rate (%)*. Total Cost calculates automatically.
4. **Payment Recording:** Input *Paid Amount (Rs.)*. The system computes *Remaining Balance* and updates the vendor's payable ledger.
5. **Stock Inbound:** Click the green **Save & Receive Stock** button. The inventory increments atomically via `PurchaseReceive` movements, and goods are immediately available at the POS terminal.

09 Supplier Directory & Accounts Payable

NAVIGATION Click 🚚 **Suppliers** on the left navigation sidebar.

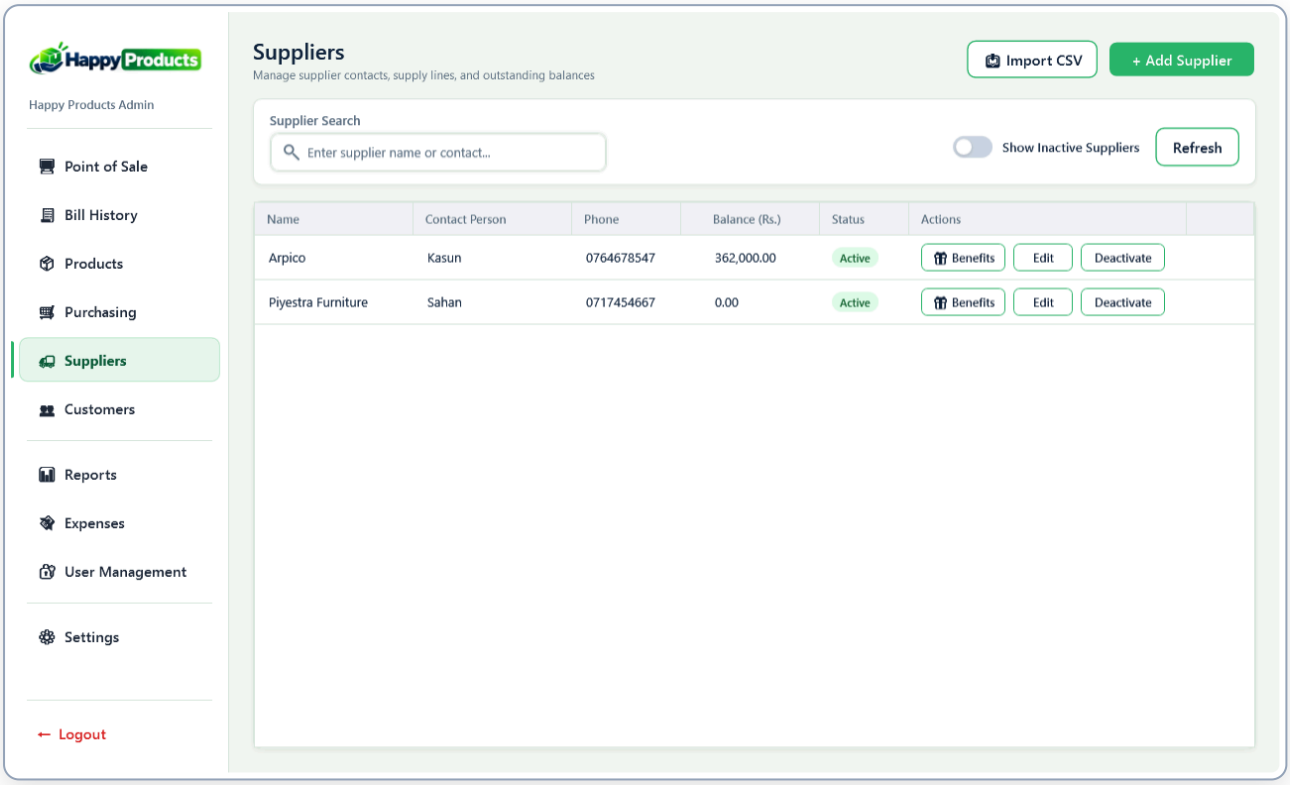


Figure 9.1: Actual Suppliers screen displaying vendor contacts, outstanding payable balances, and benefit actions.

9.1 Managing Vendors & Balances

- **Master Directory:** Displays Supplier Name (e.g., Arpico, Piyestra Furniture), Contact Person, Phone number, and current **Balance (Rs.)** owed to the supplier.
- **Adding Vendors:** Click the green **+ Add Supplier** button to register a new vendor company, representative, and address.
- **Supplier Benefits & Entitlements:** Click **Benefits** on any supplier row to view contractual free items, special trade discounts, or rebate agreements.
- **Import CSV:** Batch import supplier lists using the **Import CSV** tool.
- **Settling Accounts:** When payments are made by cheque or bank transfer, record payment entries to decrement the supplier's outstanding balance.

10 Customer Directory & Credit Management

NAVIGATION Click 👥 **Customers** on the left navigation sidebar.

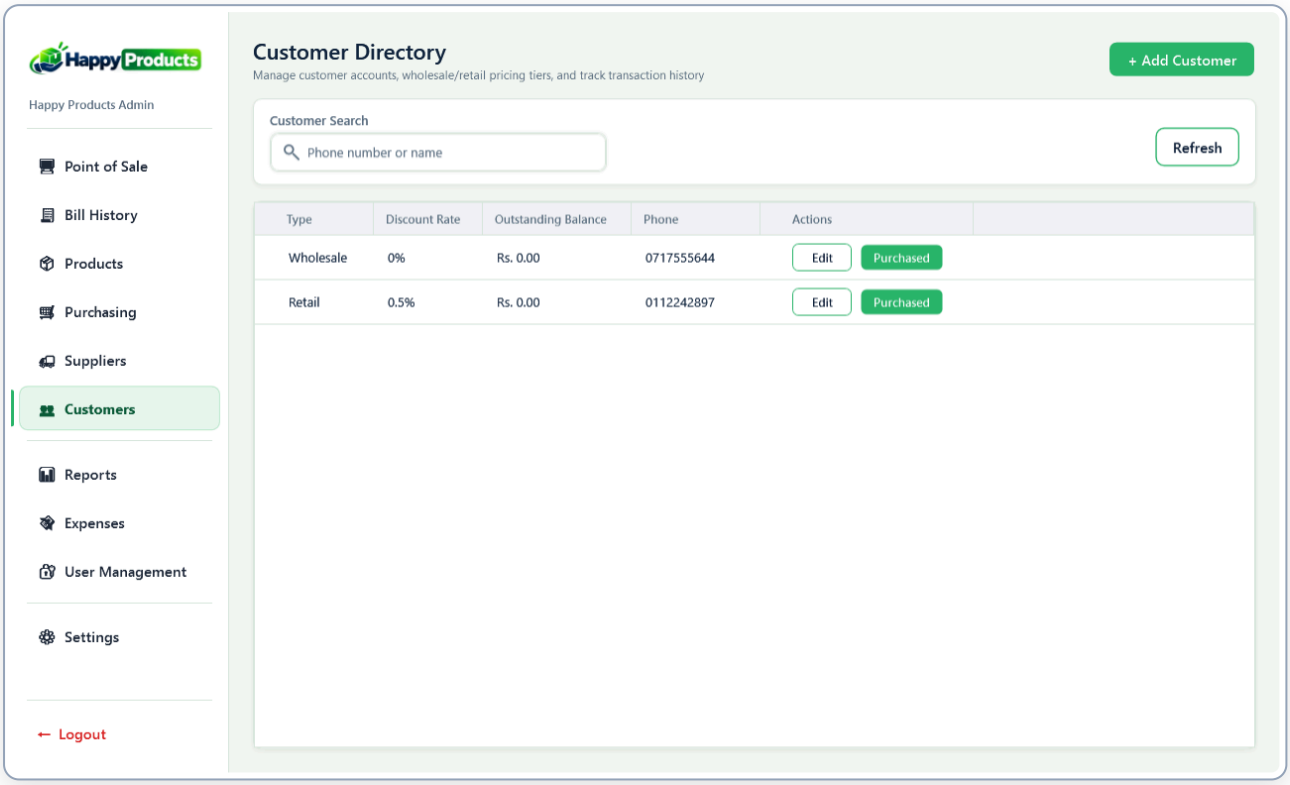


Figure 10.1: Actual Customer Directory showing wholesale/retail tiers, discount rates, balances, and purchase lookups.

10.1 Customer Directory Capabilities

- **Customer Search:** Type customer name or phone number into the search bar to locate accounts in under 1 second.
- **Account Types & Discounts:** Customers are tagged as *Retail* or *Wholesale* with preset default discount rates (e.g., 0.5%).
- **Outstanding Balance Tracking:** Displays client credit balances in real time.
- **Purchased History:** Click the green **Purchased** button on any row to open the client's lifetime purchase history, verifying past invoices, items, and warranty dates.
- **Adding Customers:** Click **+ Add Customer** to record new commercial contractors, setting credit limits and phone numbers.

11 Warehouse Operational Expenses

NAVIGATION Click **Expenses** on the left navigation sidebar (Admin access required).

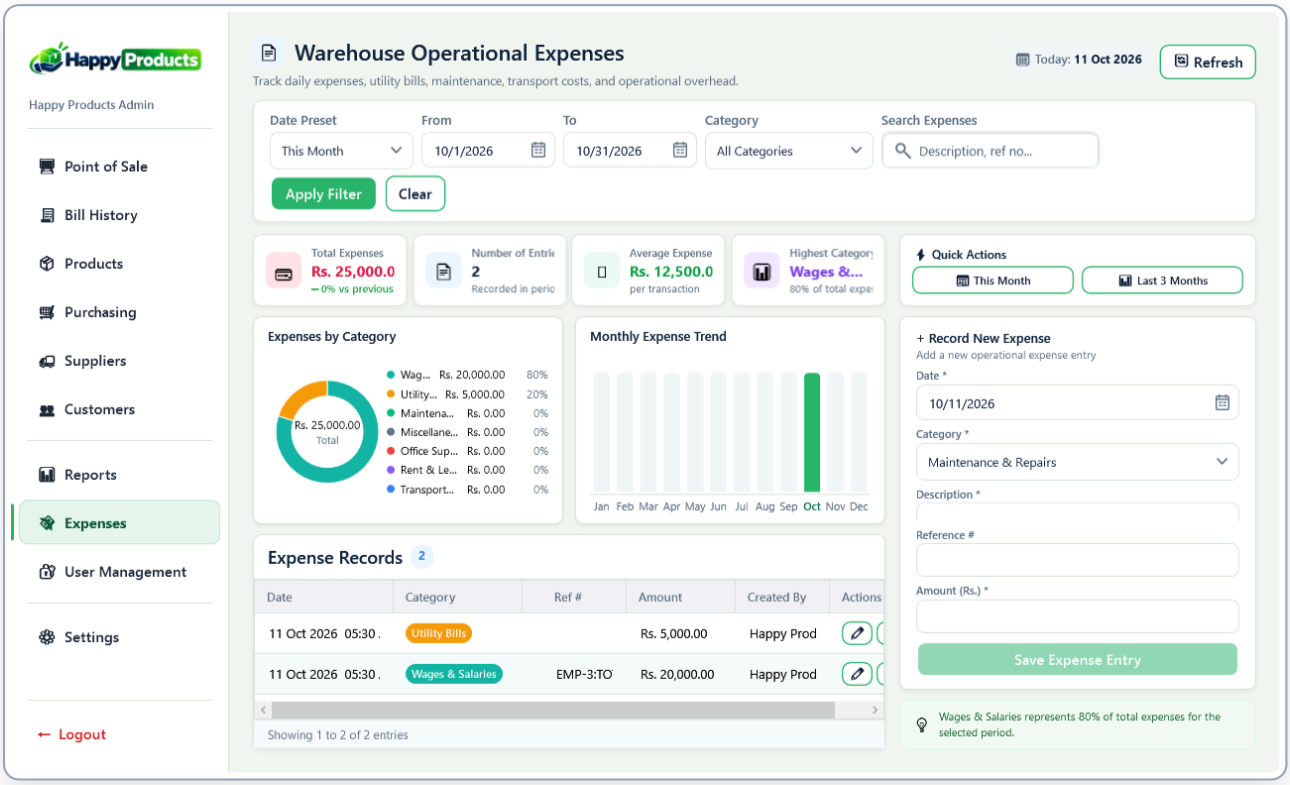


Figure 11.1: Actual Warehouse Operational Expenses view with category breakdowns, monthly trend charts, and expense logging.

11.1 Logging & Analyzing Operational Overhead

- **KPI Summary Cards:** Displays *Total Expenses (Rs.)*, *Number of Entries*, *Average Expense (Rs.)*, and *Highest Category*.
- **Interactive Charts:** Visual donut chart illustrating *Expenses by Category* (Wages & Salaries, Utility Bills, Transport & Fuel, Rent, etc.) and a 12-month *Monthly Expense Trend* bar chart.
- **Recording an Expense:** Use the right-hand form: Select *Date*, pick *Category* (Utility Bills, Maintenance & Repairs, Transport, etc.), enter *Description*, optional *Reference #*, input *Amount (Rs.)*, and click **Save Expense Entry**.
- **Audit Ledger:** The bottom table displays all historical expense entries, recording the user who authorized the payment.

12 Reports & Executive Analytics

NAVIGATION Click **Reports** on the left navigation sidebar (Admin access required).

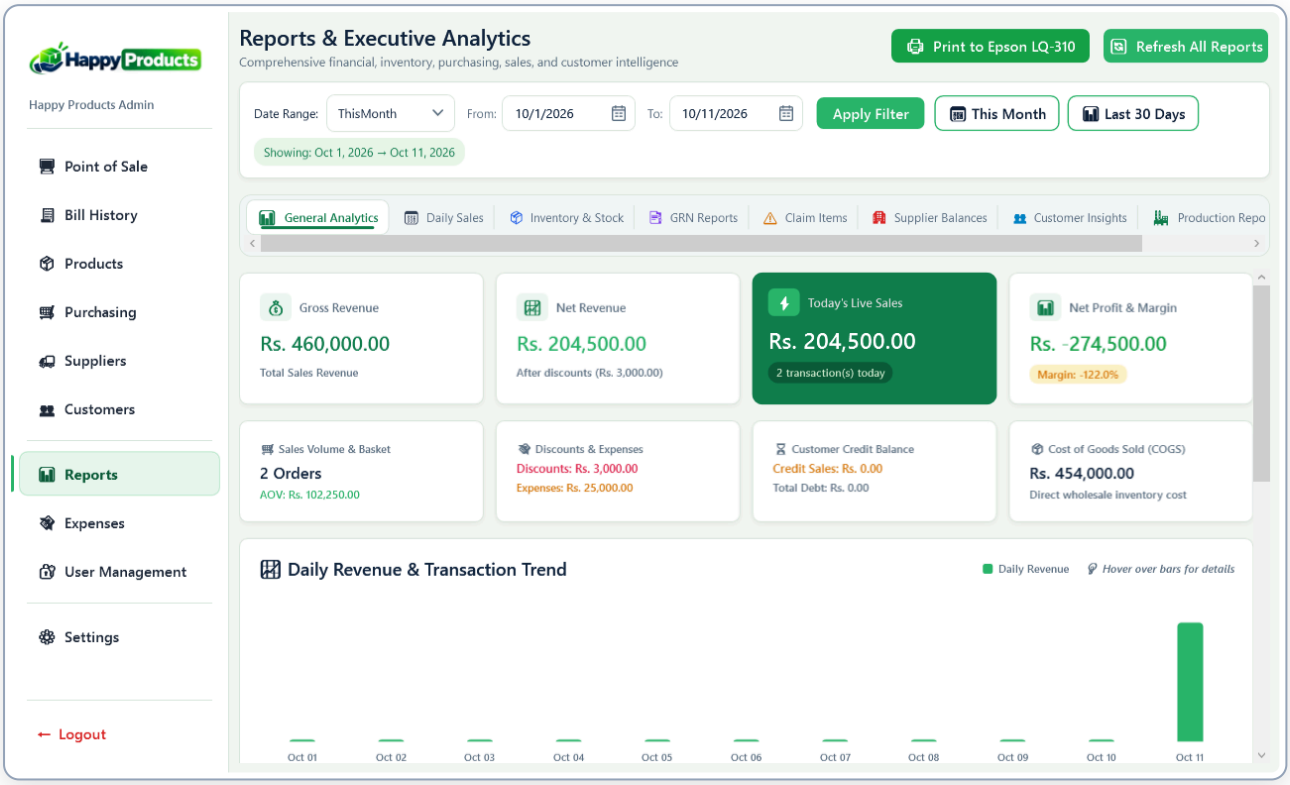


Figure 12.1: Actual Reports & Executive Analytics dashboard showing real-time revenue cards, trend charts, and report tabs.

12.1 Analytical Capabilities & Report Tabs

- **Executive KPI Cards:** Real-time figures for *Gross Revenue (Rs.)*, *Net Revenue (Rs.)*, *Today's Live Sales (Rs.)*, *Net Profit & Margin*, *Sales Volume*, *Discounts & Expenses*, and *Cost of Goods Sold (COGS)*.
- **Interactive Trend Chart:** Graphical daily revenue distribution chart showing transaction trends across the month.
- **Dedicated Intelligence Tabs:**
  - **General Analytics:** High-level financial overview and margins.
  - **Daily Sales:** Shift-by-shift sales totals and counter receipts.
  - **Inventory & Stock:** Inventory valuation (Cost vs Retail) and stock reorder reports.
  - **GRN Reports:** Inbound purchasing history and supplier goods receipts.
  - **Supplier Balances:** Summary of outstanding payables across all vendors.
  - **Customer Insights:** Top buyers, aging receivables, and credit ledger.
- **Hardware Printing:** Click **Print to Epson LQ-310** to print the active report sheet on continuous tractor paper.

13 User Management & Payroll Setup

NAVIGATION Click 🔐 **User Management** on the left navigation sidebar (Admin access required).

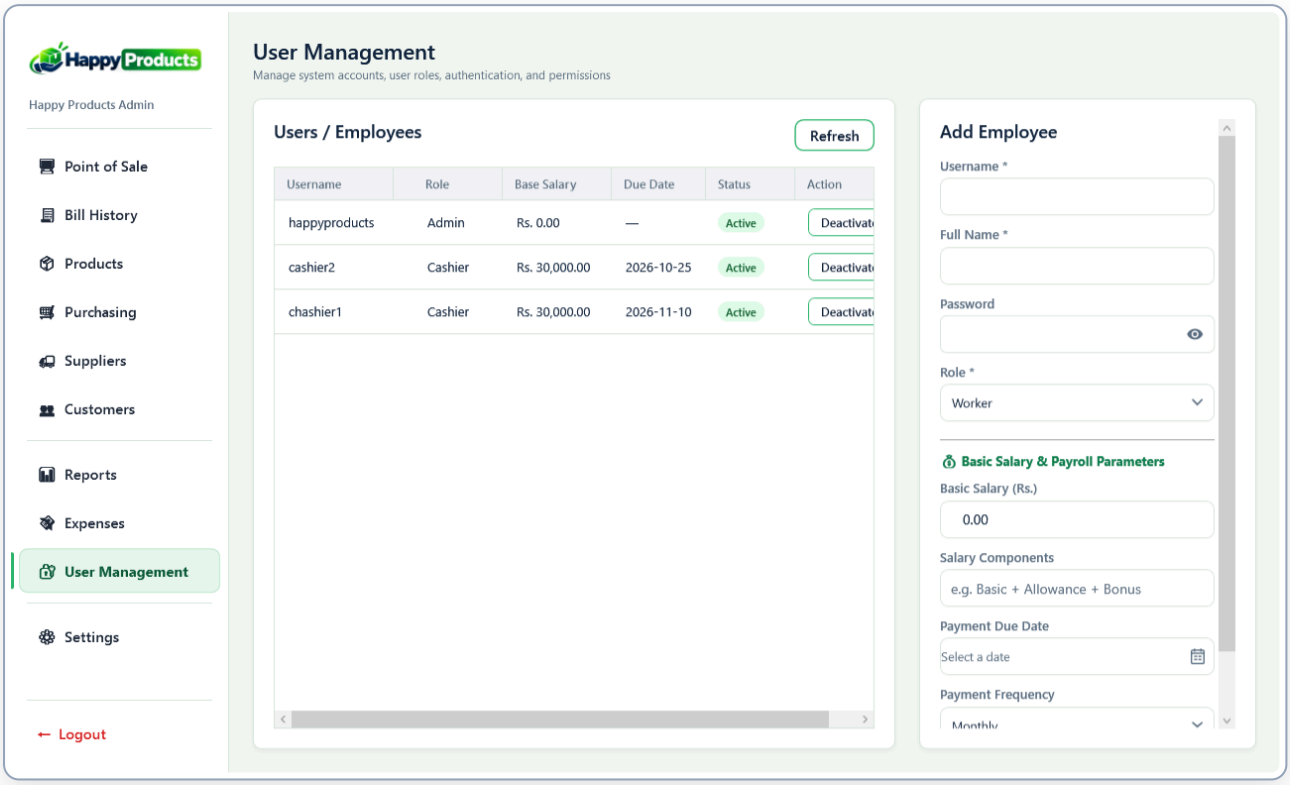


Figure 13.1: Actual User Management view showing staff directory, roles, base salaries, and add employee form.

13.1 Managing Staff Accounts & Permissions

- **Employee Directory:** Lists all staff accounts (e.g., `happyproducts`, `cashier1`, `cashier2`), showing assigned Role (*Admin*, *Cashier*, *Worker*), Base Salary (Rs.), Payment Due Date, Status (*Active*), and Deactivate action.
- **Adding New Staff Members:**
  1. Enter unique *Username* and *Full Name* in the right-hand panel.
  2. Set temporary *Password* (automatically salted and hashed via BCrypt).
  3. Select *Role* (`WORKER`, `ADMIN`).
  4. Configure payroll: Set *Basic Salary (Rs.)*, optional *Salary Components*, *Payment Due Date*, and *Payment Frequency* (Monthly).
  5. Click **Save User** to activate the account.
- **Safe Deactivation:** Click **Deactivate** on an employee row to immediately revoke login access while preserving all historical sales records and audit logs.

14 System Settings, Printer & Backup Maintenance

NAVIGATION Click ⚙ **Settings** on the left navigation sidebar (Admin access required).

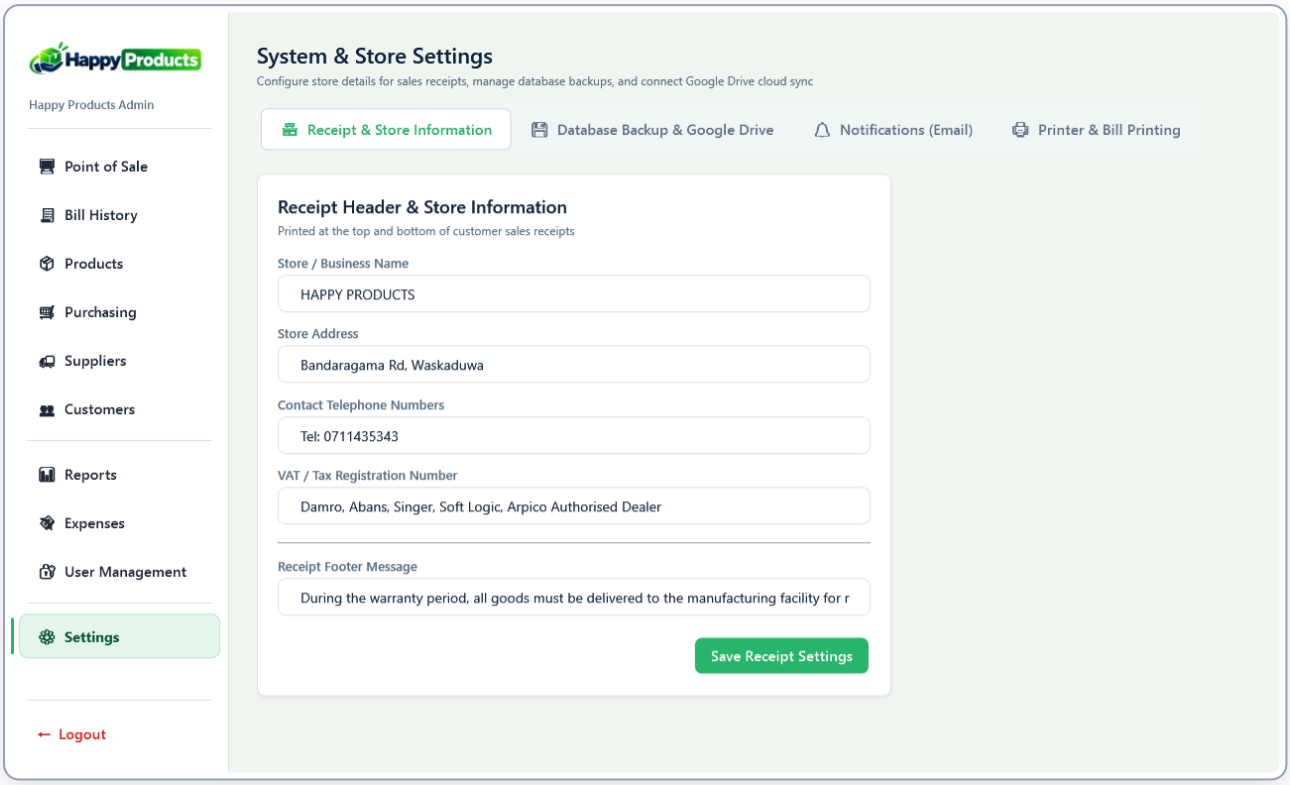


Figure 14.1: Actual System & Store Settings screen showing receipt headers, backup controls, and printer settings.

14.1 Configuration Modules

- **Receipt & Store Information Tab:**
  - *Store / Business Name:* `HAPPY PRODUCTS`
  - *Store Address:* `Bandaragama Rd, Waskaduwa`
  - *Contact Telephone Numbers:* `Tel: 0711435343`
  - *VAT / Tax Registration Number:* `Damro, Abans, Singer, Soft Logic, Arpico Authorised Dealer`
  - *Receipt Footer Message:* Warranty policies and goods return disclaimer clauses.
  - Click **Save Receipt Settings** to update receipt print headers immediately.
- **Database Backup & Google Drive Tab:** Trigger on-demand manual database backups. Backups are stored in `C:\ProgramData\WarehousePOS\Backups\`. Periodic USB flash drive export is strongly recommended.
- **Printer & Bill Printing Tab:** Configure default Windows spooler printer (`EPSON LQ-310 ESC/P2`), set continuous 5.5-inch page format, and calibrate feed lines.

15 Daily Checklists & Shortcut Hotkeys

15.1 Daily Cashier Opening Checklist

1. **Boot Workstation:** Power on host PC. Verify system date and time are accurate.
2. **Power Epson LQ-310:** Verify green *Ready* LED is illuminated. Ensure continuous 2-ply tractor paper is aligned at perforation line.
3. **Launch POS & Login:** Open WarehousePOS, input individual credentials, and sign in.
4. **Count Cash Float:** Verify opening cash drawer balance matches opening register records.

15.2 Daily Store Closing Checklist

1. **Reconcile Daily Register:** Navigate to **Reports** ➔ *Today's Live Sales*. Balance cash and card slips against cash drawer contents.
2. **Print Closing Summary:** Click **Print to Epson LQ-310** to generate the signed daily audit slip.
3. **Trigger Database Backup:** Navigate to **Settings** ➔ *Database Backup*, click **Backup Database Now**. Copy to external USB drive weekly.
4. **Logout & Shutdown:** Click **Logout** on the sidebar and shut down Windows cleanly.

15.3 Complete Keyboard Shortcut Reference

Shortcut Key	Action Performed	Context / Screen
`F1`	Focus Product Search Bar	Point of Sale (POS)
`F2`	Focus Customer Search Input	Point of Sale (POS)
`F4`	Toggle Retail / Wholesale Price	Point of Sale (POS)
`F8`	Focus Discount Input Box	Point of Sale (POS)
`F9`	Focus Cash Amount Paid Box	Point of Sale (POS)
`F12` or `Ctrl` + `Enter`	Complete Sale & Print Bill	Point of Sale (POS)
`Ctrl` + `P`	Reprint Last Receipt	POS / Bill History
`Esc`	Clear Search / Close Modal	All Application Windows

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